

Module-3 (Assignment Day 4)

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

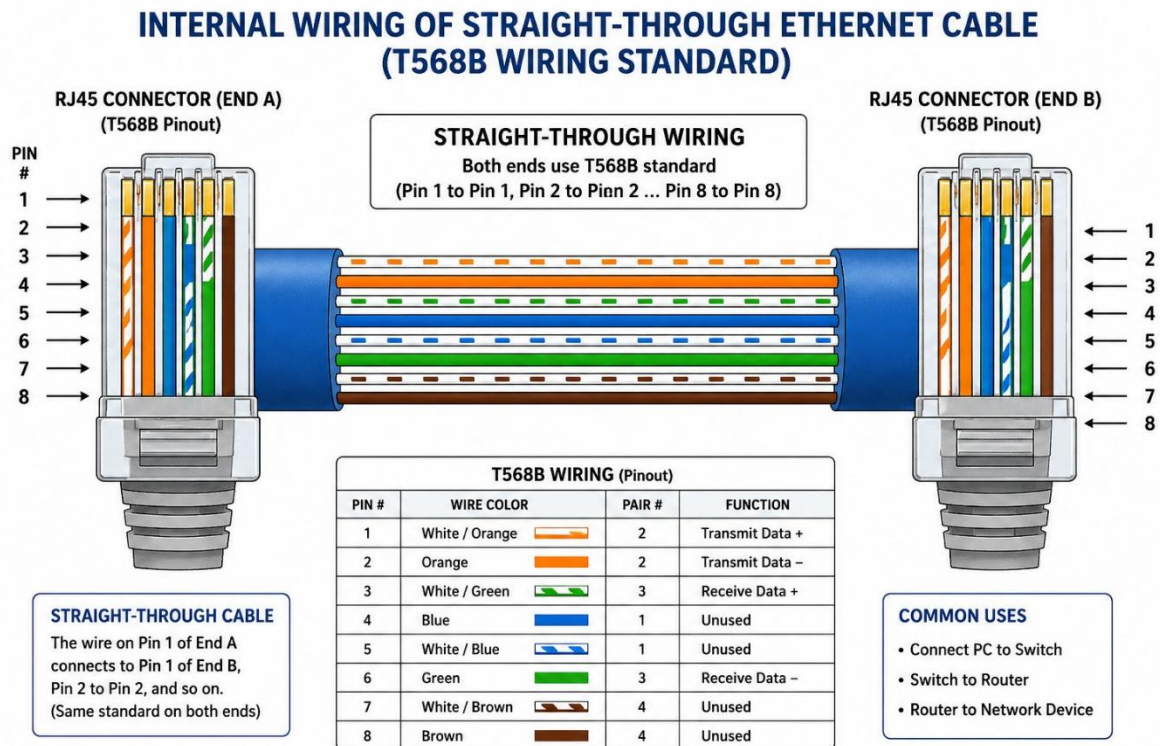


Difference	Ethernet Cable	Fiber Optic Cable
1. Signal type	Uses electrical signals through copper wires.	Uses light signals through glass or plastic fibers.
2. Speed & distance	Good for shorter distances, usually up to about 100 m per copper Ethernet link.	Supports very high speeds over much longer distances.
3. Cost & durability	Cheaper and easier to install.	More expensive and requires more careful handling.

Real-world examples:

- **Ethernet:** Connecting a gaming PC to a **home Wi-Fi router** using a LAN cable.
- **Fiber optic:** Connecting buildings in a **college campus backbone** for high-speed communication.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.



Step-by-step: Attach an RJ45 connector to a CAT6 cable:-

1. **Cut the cable** – Cut the CAT6 cable to the required length.
2. **Strip the outer cover** – Remove about 2–3 cm (around 1 inch) of the outer jacket. Be careful not to damage the small wires inside.
3. **Separate the wires** – Untwist the four pairs and straighten the eight individual wires. Keep the untwisted section as short as practical.
4. **Arrange the wires in T568B order** – From left to right, arrange them as:
White-Orange -> Orange -> White-Green -> Blue -> White-Blue -> Green -> White-Brown -> Brown.
5. **Trim the wires** – Hold the wires tightly in the correct order and cut their ends evenly so they can enter the connector together.
6. **Insert into the RJ45 plug** – Push all eight wires into the connector. Make sure each wire reaches the front of the plug and the cable jacket goes inside the connector far enough for the strain relief.
7. **Check the order** – Before crimping, look through the clear connector and confirm the colors are still in the correct T568B order.
8. **Crimp the connector** – Place the RJ45 plug into the correct slot of the crimping tool and squeeze the handles firmly until the connector is secured.
9. **Repeat on the other end** – For a straight-through cable, use the same T568B order on both ends.
10. **Test the cable** – Connect both ends to a LAN cable tester. The tester should show the pins in the correct sequence. If there is a wiring error, cut off the connector and redo it.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.



For a gaming café with 10 PCs, I would choose Ethernet (CAT6) cables because they provide enough speed for LAN gaming and are cheaper than fiber optics. I would use an RJ45 crimper to attach RJ45 connectors to the Ethernet cables. Since the PCs are usually close to the network switch, Ethernet is suitable for the required distance. Fiber optics and a fiber cleaver would be more expensive and are mainly useful for longer distances or high-speed backbone connections.